22 Find, algebraically, the set of values of x such that

$$x^2 < 25 \quad \textbf{and} \quad 4x(x-1) > 3$$

$x^2 < 25$ gives $-5 < x < 5$.

1 mark
Bound x on BOTH sides.

$4x(x - 1) > 3$ gives $4x^2 - 4x - 3 > 0$.

Factorise: $(2x + 1)(2x - 3) > 0$.

1 mark
Expand, move 3 to the left, then factorise.

Critical values: $x = -0.5$ and $x = 1.5$.

1 mark
Keep all four critical endpoints.

The product is positive outside the two roots:

$x < -0.5$ or $x > 1.5$.

1 mark
Outside the roots, both factors have the same sign.

Intersect these with $-5 < x < 5$:

$-5 < x < -0.5$ or $1.5 < x < 5$.

1 mark
Both original inequalities must hold; all endpoints are excluded.

$-5 < x < -0.5$ or $1.5 < x < 5$

(Total for Question 22 is 5 marks)

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